

Privacy in Statistics and Machine Learning
In-class Exercises for Lecture 17 (Multiplicative Weights)
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Problems with marked with an asterisk () are more challenging or open-ended.*

1. **(Online Learning Requires Randomization)** Show that for every method $\mathcal{D}$ that plays deterministic actions (where $\mathbf{p}^t$ puts probability 1 on a single action) there is an adversary for which $\mathcal{D}$'s average regret is $\Omega(1)$.
2. **($\sqrt{\ln(k)/T}$ lower bound)** Show that for any method (randomized or not), if the adversary picks cost vectors uniformly at random in $\{0, 1\}^k$, the expected regret will be $\Theta(\sqrt{\ln(k)/T})$. (*Hint:* The expected cost paid by the algorithm is exactly $T/2$. Show that in hindsight, with probability at least $1/2$, one of the choices will have cost less than $\frac{T}{2} - \Omega(\sqrt{T \ln(k)})$. Start by showing $\frac{T}{2} - \Omega(\sqrt{T})$.)
This means that the guarantee obtained by multiplicative weights is tight, in general.
3. **(MW with a perfect action)** Suppose we know ahead of time that there is a perfect choice a^* that always has cost 0. Show that we can set η so that the algorithm achieves total expected cost at most $2 \ln(k)$. (Be careful: our proof required η to be at most $1/2$.)
4. **(*)** Show that if we know OPT ahead of time, we can set η to get an expected average cost of at most $\frac{OPT}{T} + \frac{2}{T} \sqrt{\ln(k) \cdot \max(OPT, \ln(k))}$.
5. Theorem 3.1 requires us to know the number of steps T ahead of time. Show that one can modify the algorithm to adapt automatically to the length of the process. Specifically, there is a standard trick known as “repeated doubling”: we start the algorithm assuming we will run for $T_0 = 4 \ln(k)$ steps. If the number of steps exceeds T_0 , we restart the algorithm assuming a length of $T_1 = 2T_0$. If the number of steps exceeds $T_0 + T_1$, we expand our time horizon to $T_2 = 2T_1$, and so on. Show that this variation achieves average regret $O(\sqrt{\ln(k)/T})$ of T (without knowing T).
6. How important is it to select higher-cost actions with exponentially small probability? Consider an algorithm that, at each time t , selects action a with probability that scales polynomially in its cost so far $c_a^{<t}$. Specifically, suppose $p_a^t \propto \frac{1}{1+c_a^{<t}}$. Show a sequence of cost vectors on which the algorithm has expected average regret at least $\Omega(k/T)$ (one can prove a stronger bound; but even this bound highlights the bad dependency on k).